Food Delivery Platform

Database Design Document — Final Aligned

Version 2.0 I PostgreSQL 15+ with PostGIS I 49 API endpoints across 6 microservices I Scale: 50M users, 1M restaurants, 1M drivers, 2M DAU

Table of Contents

1. Prerequisites & Extensions

```
CREATE EXTENSION IF NOT EXISTS "uuid-ossp";   -- UUID generation
CREATE EXTENSION IF NOT EXISTS "pgcrypto";    -- digest, gen_random_uuid, hashing
CREATE EXTENSION IF NOT EXISTS "postgis";     -- ST_DWithin used by restaurant discovery (Redis GEO is cache only)
CREATE EXTENSION IF NOT EXISTS "pg_partman";  -- partition management
```

2. Enum Types

```
CREATE TYPE user_role           AS ENUM ('client','restaurant_owner','restaurant_manager','driver');
CREATE TYPE account_status      AS ENUM ('active','suspended','deleted');
CREATE TYPE gender_type         AS ENUM ('male','female','other');
CREATE TYPE onboarding_status   AS ENUM ('draft','pending_verification','in_review','approved','rejected');

-- document_type: REPLACED with VARCHAR(50) referencing document_type_definitions (CHANGE #18)

CREATE TYPE upload_status       AS ENUM ('pending','uploaded','verified','rejected');

-- vehicle_type aligned with LLD canonical taxonomy (CHANGE #1)
CREATE TYPE vehicle_type        AS ENUM ('bike','scooter','car');

CREATE TYPE platform_type       AS ENUM ('android','ios','web');
CREATE TYPE restaurant_status   AS ENUM ('verification_pending','pending_review','active','suspended','rejected');

-- order_status aligned with LLD timeline (CHANGE #2)
```

```
CREATE TYPE order_status AS ENUM (
  'order_created','pending_payment','confirmed','accepted','preparing',
  'ready_for_pickup','picked_up','out_for_delivery','delivered','cancelled','rejected'
);

CREATE TYPE payment_method     AS ENUM ('upi','card','wallet','cod','net_banking');
CREATE TYPE payment_status     AS ENUM ('pending','authorized','captured','failed','refunded');
CREATE TYPE gateway_name       AS ENUM ('razorpay','cashfree','stripe','paytm');
CREATE TYPE refund_status      AS ENUM ('initiated','processed','failed','rejected');

CREATE TYPE delivery_request_status AS ENUM ('pending_acceptance','accepted','rejected','expired','no_partners');
CREATE TYPE delivery_status       AS ENUM ('assigned','en_route_to_restaurant','arrived_at_restaurant','picked_up','out_
for_delivery','delivered');

-- assignment_strategy: REPLACED with VARCHAR(32) + app-side whitelist (CHANGE #16)

CREATE TYPE recipient_type     AS ENUM ('client','restaurant_owner','restaurant_manager','driver');  -- CHANGE #3

-- notification priority: REPLACED with SMALLINT 1..10 (CHANGE #17)

CREATE TYPE notification_status  AS ENUM ('queued','processing','sent','delivered','failed','failed_at_queue','failed_at_delivery');
CREATE TYPE location_source     AS ENUM ('gps','network','manual');
CREATE TYPE audit_action       AS ENUM ('create','update','delete','login','logout','status_change','refund','payment');
```

3. Utility: Auto-Update Trigger and Status-History Triggers

```
-- 3.1 Auto-update updated_at on every UPDATE
CREATE OR REPLACE FUNCTION trigger_set_updated_at()
RETURNS TRIGGER AS $$
BEGIN
  NEW.updated_at = NOW();
  RETURN NEW;
END;
$$ LANGUAGE plpgsql;

-- Applied per table that has updated_at:
-- CREATE TRIGGER set_updated_at BEFORE UPDATE ON <table>
--   FOR EACH ROW EXECUTE FUNCTION trigger_set_updated_at();

-- 3.2 Order status history auto-emitter (NEW)
CREATE OR REPLACE FUNCTION trigger_emit_order_status_history()
RETURNS TRIGGER AS $$
BEGIN
  IF NEW.status IS DISTINCT FROM OLD.status THEN
    INSERT INTO order_status_history(order_id, from_status, to_status, changed_by, changed_at)
    VALUES (NEW.order_id, OLD.status, NEW.status, NEW.updated_by, NOW());
  END IF;
  RETURN NEW;
END;
$$ LANGUAGE plpgsql;

CREATE TRIGGER trg_orders_status_history AFTER UPDATE OF status ON orders
  FOR EACH ROW EXECUTE FUNCTION trigger_emit_order_status_history();

-- 3.3 Delivery status history auto-emitter (analogous; identical pattern on deliveries)
CREATE OR REPLACE FUNCTION trigger_emit_delivery_status_history()
RETURNS TRIGGER AS $$
BEGIN
  IF NEW.status IS DISTINCT FROM OLD.status THEN
    INSERT INTO delivery_status_history(delivery_id, from_status, to_status, changed_by, changed_at)
    VALUES (NEW.delivery_id, OLD.status, NEW.status, NEW.updated_by, NOW());
```

```
  END IF;
  RETURN NEW;
END;
$$ LANGUAGE plpgsql;

CREATE TRIGGER trg_deliveries_status_history AFTER UPDATE OF status ON deliveries
  FOR EACH ROW EXECUTE FUNCTION trigger_emit_delivery_status_history();
```

4. Table Definitions

Classification tags per column: [PII], [Sensitive Financial], [PCI], [Internal], [Public]. Tables with audit columns: created_at, created_by, updated_at, updated_by. Soft delete: is_deleted, deleted_at, deleted_by.

4.1 users

Purpose: Central identity table for all roles (client, restaurant_owner, restaurant_manager, driver).

Audit columns: Yes | Soft delete: Yes | APIs: 1–5, 10–11

Column	Type	Null	Default	Constraints	Class
user_id	UUID	NO	gen_random_uuid()	PK	Internal
phone	VARCHAR(15)	NO			PII
name	VARCHAR(100)	NO			PII
email	VARCHAR(255)	YES			PII
email_verified	BOOLEAN	NO	FALSE		Internal
phone_verified	BOOLEAN	NO	FALSE		Internal
role	user_role	NO			Internal
account_status	account_status	NO	'active'		Internal
profile_image_s3_key	VARCHAR(512)	YES			Internal
referral_code	VARCHAR(20)	YES			Internal
onboarding_complete	BOOLEAN	NO	FALSE		Internal
is_deleted	BOOLEAN	NO	FALSE		Internal
deleted_at	TIMESTAMPTZ	YES			Internal
deleted_by	UUID	YES			Internal
created_at	TIMESTAMPTZ	NO	NOW()		Internal
created_by	UUID	YES			Internal
updated_at	TIMESTAMPTZ	NO	NOW()		Internal
updated_by	UUID	YES			Internal

Note: is_active dropped per CHANGE #15 (redundant with account_status).

Indexes:

```
CREATE UNIQUE INDEX idx_users_phone_role ON users(phone, role) WHERE is_deleted = FALSE;
CREATE INDEX idx_users_email        ON users(email) WHERE email IS NOT NULL AND is_deleted = FALSE;
CREATE INDEX idx_users_role         ON users(role) WHERE is_deleted = FALSE;
CREATE INDEX idx_users_account_status ON users(account_status);
```

Trigger: set_updated_at

4.2 client_profiles

Purpose: Client-specific profile data (date of birth, gender). APIs: 6, 10, 11

Column	Type	Null	Default	Constraints	Class
user_id	UUID	NO		PK, FK→users(user_id) ON DELETE CASCADE	Internal
date_of_birth	DATE	YES			PII
gender	gender_type	YES			PII
created_at / updated_at	TIMESTAMPTZ	NO	NOW()		Internal
created_by / updated_by	VARCHAR	NO			Internal

Trigger: set_updated_at

4.3 restaurant_owner_profiles

Purpose: Restaurant owner KYC and compliance data. APIs: 6, 10

Column	Type	Null	Constraints	Class
user_id	UUID	NO	PK, FK→users(user_id) ON DELETE CASCADE	Internal
fssai_license_number_encrypted	BYTEA	YES		Sensitive Financial
fssai_expiry_date	DATE	YES		Internal
gstin_encrypted	BYTEA	YES		Sensitive Financial
pan_number_encrypted	BYTEA	YES		Sensitive Financial
created_at / updated_at / created_by / updated_by	…			Internal

Trigger: set_updated_at

4.4 driver_profiles

Purpose: Driver-specific data — license, vehicle, availability. APIs: 6, 10, 11, 41, 45, 46

Column	Type	Null	Default	Constraints	Class
user_id	UUID	NO		PK, FK→users CASCADE	Internal
driving_license_number_ encrypted	BYTEA	NO			Sensitive Financial
driving_license_expiry	DATE	NO			Internal
vehicle_type	vehicle_type	NO			Internal
vehicle_registration_encrypted	BYTEA	NO			Sensitive Financial
vehicle_make	VARCHAR(50)	YES			Internal
vehicle_model	VARCHAR(50)	YES			Internal
vehicle_year	SMALLINT	YES		CHECK(vehicle_year >= 2000)	Internal
insurance_expiry_date	DATE	YES			Internal
is_available	BOOLEAN	NO	FALSE		Internal
current_order_id	UUID	YES		FK→orders SET NULL	Internal
current_city	VARCHAR(80)	YES			Internal
rating	DECIMAL(3,2)	YES		CHECK(rating BETWEEN 1 AND 5)	Public
total_deliveries	INTEGER	NO	0		Public
created_at / updated_at / created_by / updated_by	…				Internal

```
CREATE INDEX idx_driver_profiles_available  ON driver_profiles(is_available) WHERE is_available = TRUE;
CREATE INDEX idx_driver_profiles_city_avail ON driver_profiles(current_city, is_available);
```

Trigger: set_updated_at

4.5 bank_accounts

Purpose: Bank details for restaurant owners and drivers (payouts). APIs: 6, 10

Column	Type	Null	Default	Constraints	Class
bank_account_id	UUID	NO	gen_random_uuid()	PK	Internal
user_id	UUID	NO		FK→users CASCADE	Internal
account_number_encrypted	BYTEA	NO			Sensitive Financial
ifsc_code	VARCHAR(11)	NO			Sensitive Financial
account_holder_name	VARCHAR(100)	NO			PII
is_primary	BOOLEAN	NO	TRUE		Internal
is_verified	BOOLEAN	NO	FALSE		Internal
created_at / updated_at	TIMESTAMPTZ	NO	NOW()		Internal

```
CREATE INDEX idx_bank_accounts_user       ON bank_accounts(user_id);
CREATE UNIQUE INDEX uq_bank_accounts_primary ON bank_accounts(user_id) WHERE is_primary = TRUE;
```

Trigger: set_updated_at

4.6 otp_requests

Purpose: OTP generation, verification, and rate-limiting. APIs: 2, 3, 4

Column	Type	Null	Default	Constraints	Class
otp_id	UUID	NO	gen_random_uuid()	PK	Internal
user_id	UUID	YES		FK→users CASCADE	Internal
phone	VARCHAR(15)	NO			PII
device_id	VARCHAR(255)	YES			Internal
ip_address	INET	YES			PII
otp_hash	VARCHAR(255)	NO		Never store plaintext	PII
expires_at	TIMESTAMPTZ	NO			Internal
is_verified	BOOLEAN	NO	FALSE		Internal
attempts	SMALLINT	NO	0	CHECK(attempts <= 5)	Internal
resend_count	SMALLINT	NO	0	CHECK(resend_count <= 5)	Internal
last_sent_at	TIMESTAMPTZ	YES			Internal
blocked_until	TIMESTAMPTZ	YES			Internal
created_at	TIMESTAMPTZ	NO	NOW()		Internal

```
-- Verify-OTP lookup: phone + device + unverified
CREATE INDEX idx_otp_verify
  ON otp_requests (phone, device_id, created_at DESC)
  WHERE is_verified = FALSE;

-- User cooldown check: last OTP sent by this user
CREATE INDEX idx_otp_user_cooldown
  ON otp_requests (user_id, last_sent_at DESC);

-- Cleanup job: delete expired rows nightly
CREATE INDEX idx_otp_expires ON otp_requests (expires_at);
```

4.7 sessions

Purpose: Server-side session store (Redis is primary; this table is for audit/persistence). APIs: 4, 5

Column	Type	Null	Default	Constraints	Class
session_id	UUID	NO	gen_random_uuid()	PK	Internal
user_id	UUID	NO		FK→users CASCADE	Internal
phone	VARCHAR(15)	NO			PII
role	user_role	NO			Internal
device_id	VARCHAR(100)	NO			Internal
is_active	BOOLEAN	NO	TRUE		Internal
logged_in_at	TIMESTAMPTZ	NO	NOW()		Internal
last_active_at	TIMESTAMPTZ	NO	NOW()		Internal
logged_out_at	TIMESTAMPTZ	YES			Internal
ip_address	INET	YES			PII
platform	platform_type	YES			Internal
active_cart_id	VARCHAR(255)	YES			Internal
active_order_id	UUID	YES			Internal
expires_at	TIMESTAMPTZ	NO			Internal

```
CREATE INDEX idx_sessions_user_active ON sessions(user_id) WHERE is_active = TRUE;
CREATE INDEX idx_sessions_device      ON sessions(user_id, device_id);
```

4.8 onboardings

Purpose: Tracks onboarding workflow per user+role. Audit columns: Yes. APIs: 6, 7, 8, 9

Column	Type	Null	Default	Constraints	Class
onboarding_id	UUID	NO	gen_random_uuid()	PK	Internal
user_id	UUID	NO		FK→users CASCADE	Internal

role	user_role	NO			Internal
status	onboarding_status	NO	'draft'		Internal
submitted_at	TIMESTAMPTZ	YES			Internal
reviewed_by	UUID	YES		FK→users SET NULL	Internal
reviewed_at	TIMESTAMPTZ	YES			Internal
audit cols	…				Internal

```
CREATE INDEX idx_onboardings_user_role ON onboardings(user_id, role);
CREATE INDEX idx_onboardings_status    ON onboardings(status) WHERE status IN ('pending_verification','in_review');
```

Trigger: set_updated_at

4.9 onboarding_documents (+ document_type_definitions)

Purpose: Per-document upload tracking within an onboarding. APIs: 6, 7, 8, 9

CHANGE #18: document_type is now VARCHAR(50) referencing the lookup table document_type_definitions, to avoid enum migrations as countries/roles add documents.

```
CREATE TABLE document_type_definitions (
  document_type      VARCHAR(50) PRIMARY KEY,
  applicable_role    user_role NOT NULL,
  applicable_country CHAR(2)   NOT NULL DEFAULT 'IN',
  is_required        BOOLEAN   NOT NULL DEFAULT TRUE,
  description        TEXT,
  created_at         TIMESTAMPTZ NOT NULL DEFAULT NOW()
);
```

Column	Type	Null	Default	Constraints	Class
document_id	UUID	NO	gen_random_uuid()	PK	Internal
onboarding_id	UUID	NO		FK→onboardings CASCADE	Internal
document_type	VARCHAR(50)	NO		FK→document_type_definitions	Internal
s3_key	VARCHAR(512)	YES			Internal
upload_status	upload_status	NO	'pending'		Internal
review_status_message	TEXT	YES			Internal
created_at / updated_at	TIMESTAMPTZ	NO	NOW()		Internal

```
CREATE INDEX idx_onboarding_docs_onboarding ON onboarding_documents(onboarding_id);
CREATE UNIQUE INDEX uq_onboarding_doc_type   ON onboarding_documents(onboarding_id, document_type);
```

Trigger: set_updated_at

4.10 addresses

Purpose: User delivery addresses. Soft delete: Yes. APIs: 6, 12, 13, 14, 15

Column	Type	Null	Default	Constraints	Class
address_id	UUID	NO	gen_random_uuid()	PK	Internal
user_id	UUID	NO		FK→users CASCADE	Internal
label	VARCHAR(50)	YES		e.g. Home, Work	Internal
line1	VARCHAR(255)	NO			PII
line2	VARCHAR(255)	YES			PII
area	VARCHAR(100)	YES			Internal
city	VARCHAR(100)	NO			Internal
state	VARCHAR(100)	NO			Internal
pincode	VARCHAR(10)	NO			Internal
latitude	DECIMAL(9,6)	YES			Internal
longitude	DECIMAL(9,6)	YES			Internal
contact_name	VARCHAR(100)	YES			PII
contact_phone	VARCHAR(15)	YES			PII
is_default	BOOLEAN	NO	FALSE		Internal
soft delete + audit cols	…				Internal

```
CREATE INDEX idx_addresses_user ON addresses(user_id) WHERE is_deleted = FALSE;
CREATE UNIQUE INDEX uq_addresses_default ON addresses(user_id) WHERE is_default = TRUE AND is_deleted = FALSE;
```

Trigger: set_updated_at

4.11 user_locations

Purpose: Latest resolved location per user (serviceability checks). APIs: 48

Column	Type	Null	Default	Constraints	Class
user_id	UUID	NO		PK, FK→users CASCADE	Internal
latitude	DECIMAL(9,6)	NO			PII
longitude	DECIMAL(9,6)	NO			PII
accuracy_meters	REAL	YES			Internal
source	location_source	NO			Internal
captured_at	TIMESTAMPTZ	NO	NOW()		Internal

4.12 restaurants

Purpose: Restaurant profile, location, compliance, operational data. Audit/Soft-delete: Yes. APIs: 16, 17, 19, 20

Column	Type	Null	Default	Constraints	Class
restaurant_id	UUID	NO	gen_random_uuid()	PK	Internal
owner_id	UUID	NO		FK→users RESTRICT	Internal
name	VARCHAR(200)	NO			Public
restaurant_profile_image_s3_key	VARCHAR(512)	YES			Public
cuisine_types	TEXT[]	YES	'{}'		Public
rating	DECIMAL(3,2)	YES		CHECK(rating BETWEEN 1.0 AND 5.0)	Public
total_ratings	INTEGER	NO	0		Public
address_line1	VARCHAR(255)	NO			Public
city / state / pincode	VARCHAR	NO			Public
latitude / longitude	DECIMAL(9,6)	NO			Public
location	GEOMETRY(Point, 4326)	NO		PostGIS POINT	Public
service_radius_km	DECIMAL(5,2)	NO	5.0	CHECK(service_radius_km > 0)	Public
delivery_time_min	SMALLINT	YES	30		Public
opening_cron	VARCHAR(100)	YES			Public
is_open	BOOLEAN	NO	FALSE		Public
fssai_license_encrypted	BYTEA	YES			Sensitive Financial
gstin_encrypted	BYTEA	YES			Sensitive Financial
logo_s3_key	VARCHAR(512)	YES			Public
manager_name	VARCHAR(100)	YES			Internal
manager_phone	VARCHAR(15)	YES			PII
status	restaurant_status	NO	'pending_review'		Internal
soft delete + audit cols	…				Internal

```
CREATE INDEX idx_restaurants_location     ON restaurants USING GIST(location);      -- ST_DWithin
CREATE INDEX idx_restaurants_owner        ON restaurants(owner_id);
CREATE INDEX idx_restaurants_status_active ON restaurants(status) WHERE status = 'active' AND is_deleted = FALSE;
CREATE INDEX idx_restaurants_city        ON restaurants(city) WHERE is_deleted = FALSE;
CREATE INDEX idx_restaurants_cuisine      ON restaurants USING GIN(cuisine_types);
```

Trigger: set_updated_at

4.13 menus

Purpose: Menu container per restaurant (1:1 typically). APIs: 21

Column	Type	Null	Default	Constraints	Class
menu_id	UUID	NO	gen_random_uuid()	PK	Public
restaurant_id	UUID	NO		FK→restaurants CASCADE, UNIQUE	Public
created_at / updated_at	TIMESTAMPTZ	NO	NOW()		Internal

created_by	UUID	YES			Internal

Trigger: set_updated_at

4.14 menu_categories

Purpose: Category groupings within a menu (Starters, Mains, etc.). APIs: 18, 21

Column	Type	Null	Default	Constraints	Class
category_id	UUID	NO	gen_random_uuid()	PK	Public
menu_id	UUID	NO		FK→menus CASCADE	Public
name	VARCHAR(100)	NO			Public
sort_order	SMALLINT	NO	0		Public
created_at	TIMESTAMPTZ	NO	NOW()		Internal

```
CREATE INDEX idx_menu_categories_menu  ON menu_categories(menu_id);
CREATE UNIQUE INDEX uq_menu_category_name ON menu_categories(menu_id, name);
```

4.15 menu_items

Purpose: Individual food items. Audit/Soft-delete: Yes. APIs: 18, 21, 22, 23, 24, 26

Column	Type	Null	Default	Constraints	Class
item_id	UUID	NO	gen_random_uuid()	PK	Public
restaurant_id	UUID	NO		FK→restaurants CASCADE	Public
category_id	UUID	YES		FK→menu_categories SET NULL	Public
name	VARCHAR(200)	NO			Public
description	TEXT	YES			Public
price	DECIMAL(10,2)	NO		CHECK(price > 0)	Public
image_s3_key	VARCHAR(512)	YES			Public
is_veg	BOOLEAN	NO	TRUE		Public
is_available	BOOLEAN	NO	TRUE		Public
preparation_time_min	SMALLINT	YES	15		Public
tags	TEXT[]	YES	'{}'		Public
sort_order	SMALLINT	NO	0		Public
soft delete + audit cols	…				Internal

```
CREATE INDEX idx_menu_items_restaurant ON menu_items(restaurant_id) WHERE is_deleted = FALSE;
CREATE INDEX idx_menu_items_category   ON menu_items(category_id) WHERE is_deleted = FALSE;
CREATE INDEX idx_menu_items_available  ON menu_items(restaurant_id, is_available) WHERE is_deleted = FALSE;
CREATE INDEX idx_menu_items_tags       ON menu_items USING GIN(tags);
```

Trigger: set_updated_at

4.16 carts

Purpose: Cart header — one per user+device, tied to a single restaurant. APIs: 25, 26, 27, 28

Column	Type	Null	Default	Constraints	Class
cart_id	UUID	NO	gen_random_uuid()	PK	Internal
user_id	UUID	NO		FK→users CASCADE	Internal
device_id	VARCHAR(100)	NO			Internal
restaurant_id	UUID	NO		FK→restaurants CASCADE	Internal
status	VARCHAR(20)	NO	'active'	CHECK(status IN ('active','cleared','converted'))	Internal
created_at / updated_at	TIMESTAMPTZ	NO	NOW()		Internal
expires_at	TIMESTAMPTZ	YES			Internal

```
CREATE UNIQUE INDEX uq_carts_user_device ON carts(user_id, device_id) WHERE status = 'active';
CREATE INDEX idx_carts_expires         ON carts(expires_at) WHERE status = 'active';
```

Trigger: set_updated_at. Note: external cart_token = cart.<base64(cart_id)>.<hmac>; internal storage remains a UUID PK.

4.17 cart_items

Purpose: Individual items within a cart. Hard-deleted on remove. APIs: 25, 26, 27, 28

Column	Type	Null	Default	Constraints	Class
cart_item_id	UUID	NO	gen_random_uuid()	PK	Internal
cart_id	UUID	NO		FK→carts CASCADE	Internal
item_id	UUID	NO		FK→menu_items RESTRICT	Internal
quantity	SMALLINT	NO	1	CHECK(quantity > 0)	Internal
customisations	JSONB	YES	'[]'		Internal
added_at	TIMESTAMPTZ	NO	NOW()		Internal

```
CREATE INDEX idx_cart_items_cart ON cart_items(cart_id);
```

4.18 orders

Purpose: Central order record — connects user, restaurant, delivery, payment. Audit columns: Yes. Partitioning: Range on created_at (monthly). APIs: 30, 31, 32, 33, 34, 35

Column	Type	Null	Default	Constraints	Class
order_id	UUID	NO	gen_random_uuid()	PK	Internal
user_id	UUID	NO		FK→users RESTRICT	Internal
restaurant_id	UUID	NO		FK→restaurants RESTRICT	Internal
address_id	UUID	YES		FK→addresses SET NULL	Internal
delivery_address_snapshot	JSONB	NO		Immutable copy at order time	PII
payment_method	payment_method	NO			Internal
instructions	TEXT	YES			Internal
status	order_status	NO	'order_created'		Internal
subtotal / delivery_fee / taxes / discount / total_amount	DECIMAL(10,2)	NO		CHECK(>=0); total>0	Sensitive Financial
estimated_delivery_at	TIMESTAMPTZ	YES			Internal
cancelled_at	TIMESTAMPTZ	YES			Internal
cancellation_reason	TEXT	YES			Internal
idempotency_key	VARCHAR(64)	YES		UNIQUE	Internal
audit cols	…				Internal

```
CREATE INDEX idx_orders_user          ON orders(user_id, created_at DESC);
CREATE INDEX idx_orders_restaurant     ON orders(restaurant_id, created_at DESC);
CREATE INDEX idx_orders_status        ON orders(status) WHERE status NOT IN ('delivered','cancelled','rejected');
CREATE UNIQUE INDEX uq_orders_idempotency ON orders(idempotency_key) WHERE idempotency_key IS NOT NULL;
```

Trigger: set_updated_at; trg_orders_status_history (auto-emits to order_status_history).

4.19 order_items

Purpose: Snapshot of items at order time — prices frozen. APIs: 30, 31, 32

Column	Type	Null	Default	Constraints	Class
order_item_id	UUID	NO	gen_random_uuid()	PK	Internal
order_id	UUID	NO		FK→orders CASCADE	Internal
item_id	UUID	NO		FK→menu_items RESTRICT	Internal
item_name_snapshot	VARCHAR(200)	NO		Frozen at order time	Internal
item_price_snapshot	DECIMAL(10,2)	NO		CHECK(>0); frozen	Sensitive Financial
quantity	SMALLINT	NO		CHECK(quantity > 0)	Internal
customisations_snapshot	JSONB	YES	'[]'		Internal
line_total	DECIMAL(10,2)	NO		CHECK(>=0)	Sensitive Financial

```
CREATE INDEX idx_order_items_order ON order_items(order_id);
```

4.20 order_status_history

Purpose: Full audit trail of every order status transition. APIs: 30, 33, 34. Partition: monthly on changed_at.

Column	Type	Null	Default	Constraints	Class
history_id	UUID	NO	gen_random_uuid()	PK	Internal
order_id	UUID	NO		FK→orders CASCADE	Internal
from_status	order_status	YES		NULL for initial entry	Internal
to_status	order_status	NO			Internal
changed_by	UUID	YES		FK→users SET NULL	Internal
notes	TEXT	YES			Internal
changed_at	TIMESTAMPTZ	NO	NOW()		Internal

```
CREATE INDEX idx_order_status_history_order ON order_status_history(order_id, changed_at);
```

4.21 payments

Purpose: Payment lifecycle per order. Audit columns: Yes. APIs: 36, 37

Column	Type	Null	Default	Constraints	Class
payment_id	UUID	NO	gen_random_uuid()	PK	Internal
order_id	UUID	NO		FK→orders RESTRICT	Internal
user_id	UUID	NO		FK→users RESTRICT	Internal
amount	DECIMAL(10,2)	NO		CHECK(amount > 0)	Sensitive Financial
currency	VARCHAR(3)	NO	'INR'		Internal
payment_method	payment_method	NO			Internal
payment_token_encrypted	BYTEA	YES		[PCI] never log	PCI
gateway_name	gateway_name	YES			Internal
gateway_order_id / gateway_payment_id	VARCHAR(100)	YES			Sensitive Financial
gateway_signature_verified	BOOLEAN	YES			Internal
status	payment_status	NO	'pending'		Internal
amount_paid	DECIMAL(10,2)	YES			Sensitive Financial
paid_at / expires_at	TIMESTAMPTZ	YES			Internal
redirect_url	TEXT	YES			Internal
idempotency_key	VARCHAR(64)	NO		UNIQUE	Internal
audit cols	…				Internal

```
CREATE INDEX idx_payments_order       ON payments(order_id);
CREATE UNIQUE INDEX uq_payments_idempotency ON payments(idempotency_key);
CREATE INDEX idx_payments_gateway_order ON payments(gateway_order_id);
```

Trigger: set_updated_at

4.22 refunds

Purpose: Refund tracking, separate lifecycle from payments. APIs: 33

Column	Type	Null	Default	Constraints	Class
refund_id	UUID	NO	gen_random_uuid()	PK	Internal
order_id	UUID	NO		FK→orders RESTRICT	Internal
payment_id	UUID	NO		FK→payments RESTRICT	Internal
amount	DECIMAL(10,2)	NO		CHECK(amount > 0)	Sensitive Financial
status	refund_status	NO	'initiated'		Internal
gateway_refund_id	VARCHAR(100)	YES			Sensitive Financial

reason	TEXT	YES			Internal
initiated_at	TIMESTAMPTZ	NO	NOW()		Internal
processed_at	TIMESTAMPTZ	YES			Internal

```
CREATE INDEX idx_refunds_order   ON refunds(order_id);
CREATE INDEX idx_refunds_payment ON refunds(payment_id);
```

4.23 delivery_requests

Purpose: Tracks assignment attempts before a driver accepts. APIs: 41, 42. CHANGE #16: strategy is now VARCHAR(32) (app-side whitelist) instead of an enum.

Column	Type	Null	Default	Constraints	Class
request_id	UUID	NO	gen_random_uuid()	PK	Internal
order_id	UUID	NO		FK→orders RESTRICT	Internal
partner_id	UUID	YES		FK→users SET NULL	Internal
strategy	VARCHAR(32)	NO	'nearest'	App whitelist	Internal
status	delivery_request_status	NO	'pending_acceptance'		Internal
timeout_at / responded_at	TIMESTAMPTZ	YES			Internal
created_at	TIMESTAMPTZ	NO	NOW()		Internal

```
CREATE INDEX idx_delivery_requests_order   ON delivery_requests(order_id);
CREATE INDEX idx_delivery_requests_partner ON delivery_requests(partner_id) WHERE status = 'pending_acceptance';
```

4.24 deliveries

Purpose: Active delivery record after a driver accepts. Audit columns: Yes. APIs: 42, 43, 44

Column	Type	Null	Default	Constraints	Class
delivery_id	UUID	NO	gen_random_uuid()	PK	Internal
order_id	UUID	NO		FK→orders RESTRICT, UNIQUE	Internal
partner_id	UUID	NO		FK→users RESTRICT	Internal
status	delivery_status	NO	'assigned'		Internal
estimated_pickup_at / picked_up_at	TIMESTAMPTZ	YES			Internal
estimated_delivery_at / delivered_at	TIMESTAMPTZ	YES			Internal
assigned_at	TIMESTAMPTZ	NO	NOW()		Internal
audit cols	…				Internal

```
CREATE UNIQUE INDEX uq_deliveries_order      ON deliveries(order_id);
CREATE INDEX idx_deliveries_partner_active   ON deliveries(partner_id) WHERE status NOT IN ('delivered');
```

Trigger: set_updated_at; trg_deliveries_status_history.

4.25 delivery_status_history

Purpose: Full audit trail of delivery status transitions. APIs: 44

Column	Type	Null	Default	Constraints	Class
history_id	UUID	NO	gen_random_uuid()	PK	Internal
delivery_id	UUID	NO		FK→deliveries CASCADE	Internal
from_status	delivery_status	YES			Internal
to_status	delivery_status	NO			Internal
changed_by	UUID	YES		FK→users SET NULL	Internal
notes	TEXT	YES			Internal
changed_at	TIMESTAMPTZ	NO	NOW()		Internal

```
CREATE INDEX idx_delivery_status_history_delivery ON delivery_status_history(delivery_id, changed_at);
```

4.26 driver_location_logs

Purpose: High-frequency GPS location writes. Partition: daily on recorded_at. APIs: 45

Column	Type	Null	Default	Constraints	Class
log_id	UUID	NO	gen_random_uuid()	PK	Internal
partner_id	UUID	NO		FK→users CASCADE	Internal
order_id	UUID	YES		FK→orders SET NULL	Internal
latitude	DECIMAL(9,6)	NO			PII
longitude	DECIMAL(9,6)	NO			PII
bearing	REAL	YES			Internal
speed_kmh	REAL	YES			Internal
recorded_at	TIMESTAMPTZ	NO	NOW()		Internal

```
CREATE INDEX idx_driver_location_partner_time ON driver_location_logs(partner_id, recorded_at DESC);
CREATE INDEX idx_driver_location_order     ON driver_location_logs(order_id) WHERE order_id IS NOT NULL;
```

Note: ~28M rows/day at peak (1M drivers × ~28 updates/day average). Hot 48 h, cold 30 d, then delete.

4.27 notifications

Purpose: In-app notification records. Audit columns: Yes. APIs: 38, 39, 40. CHANGE #17: priority is now SMALLINT 1..10 instead of an enum.

Column	Type	Null	Default	Constraints	Class
notification_id	UUID	NO	gen_random_uuid()	PK	Internal
recipient_id	UUID	NO		FK→users CASCADE	Internal
recipient_type	recipient_type	NO			Internal
channels	TEXT[]	NO		e.g. {push,sms}	Internal
title	VARCHAR(255)	NO			PII
body	TEXT	NO			PII
data	JSONB	YES		deep_link, order_id	Internal
priority	SMALLINT	NO	5	CHECK(priority BETWEEN 1 AND 10)	Internal
status	notification_status	NO	'queued'		Internal
is_read	BOOLEAN	NO	FALSE		Internal
read_at	TIMESTAMPTZ	YES			Internal
audit cols	…				Internal

```
CREATE INDEX idx_notifications_recipient ON notifications(recipient_id, created_at DESC);
CREATE INDEX idx_notifications_unread    ON notifications(recipient_id) WHERE is_read = FALSE;
```

Trigger: set_updated_at

4.28 notification_delivery_logs

Purpose: Per-channel delivery tracking. APIs: 38

Column	Type	Null	Default	Constraints	Class
log_id	UUID	NO	gen_random_uuid()	PK	Internal
notification_id	UUID	NO		FK→notifications CASCADE	Internal
channel	VARCHAR(10)	NO		e.g. push, sms, email	Internal
provider	VARCHAR(50)	YES		e.g. FCM, Twilio, SES	Internal
provider_message_id	VARCHAR(255)	YES			Internal
status	notification_status	NO			Internal
sent_at / delivered_at	TIMESTAMPTZ	YES			Internal
failed_reason	TEXT	YES			Internal

```
CREATE INDEX idx_notif_delivery_notif ON notification_delivery_logs(notification_id);
```

4.29 audit_logs (NEW — CHANGE #19)

Purpose: Append-only audit trail (actor, action, entity, before/after, ip, ua). Partition: monthly on occurred_at. Retention: 7 years. Privileges: only audit_writer has INSERT; UPDATE/DELETE revoked from all roles.

Column	Type	Null	Default	Constraints	Class
audit_id	BIGSERIAL	NO		PK	Internal
actor_id	UUID	YES		FK→users SET NULL	Internal
actor_role	user_role	YES			Internal
action	audit_action	NO			Internal
entity_type	VARCHAR(50)	NO			Internal
entity_id	VARCHAR(64)	NO			Internal
before / after	JSONB	YES			Internal
ip_address	INET	YES			PII
user_agent	TEXT	YES			Internal
occurred_at	TIMESTAMPTZ	NO	NOW()		Internal

```
CREATE INDEX idx_audit_actor_time ON audit_logs(actor_id, occurred_at DESC);
CREATE INDEX idx_audit_entity     ON audit_logs(entity_type, entity_id, occurred_at DESC);
REVOKE UPDATE, DELETE ON audit_logs FROM PUBLIC;
```

5. Entity Relationship Diagram

flowcharts/er_diagram.svg

Mermaid source (renderable to SVG via mermaid-cli):

```
erDiagram
  users ||--o| client_profiles : has
  users ||--o| restaurant_owner_profiles : has
  users ||--o| driver_profiles : has
  users ||--o{ bank_accounts : owns
  users ||--o{ addresses : has
  users ||--o{ otp_requests : requests
  users ||--o{ sessions : opens
  users ||--o{ onboardings : starts
  onboardings ||--o{ onboarding_documents : contains
  document_type_definitions ||--o{ onboarding_documents : classifies
  users ||--o{ restaurants : owns
  restaurants ||--|| menus : has
  menus ||--o{ menu_categories : groups
  menu_categories ||--o{ menu_items : contains
  restaurants ||--o{ menu_items : sells
  users ||--o{ carts : has
  carts ||--o{ cart_items : contains
  menu_items ||--o{ cart_items : referenced_by
  users ||--o{ orders : places
  restaurants ||--o{ orders : receives
  addresses ||--o{ orders : ships_to
  orders ||--o{ order_items : contains
  menu_items ||--o{ order_items : snapshotted_in
  orders ||--o{ order_status_history : transitions
  orders ||--|| payments : paid_by
  payments ||--o{ refunds : refunded_by
  orders ||--o{ delivery_requests : assigns
  users ||--o{ delivery_requests : driver_offered
  orders ||--|| deliveries : fulfilled_by
  users ||--o{ deliveries : driver_for
```

```
deliveries ||--o{ delivery_status_history : transitions
users ||--o{ driver_location_logs : streams
users ||--o{ user_locations : last_known
users ||--o{ notifications : receives
notifications ||--o{ notification_delivery_logs : per_channel
users ||--o{ audit_logs : actor
```

6. Redis Data Structures

Key Pattern	Type	TTL	Purpose	Invalidation
session:{sid}	Hash	24h sliding	Live session state: user_id, role, device_id, is_active, active_cart_id, active_order_id, ip_address, platform	On logout / password change / role change
otp:{phone}	String	10 min	Hashed OTP for verification	On verify success
otp:rate:{phone}	Counter	1 h	Rate limit ≤5/hr	TTL only
cart:{cart_id}	Hash	24 h	Hot cart data: restaurant_id, items (JSON), updated_at	On any cart write or order placement
restaurants:geo:{city}	GEO sorted set	60 s	Discovery cache (GEORADIUS)	On restaurant create/update/status change
menu:{restaurant_id}	Hash / String JSON	5 min	Menu snapshot	On menu/item write
order:{order_id}	Hash	1 h	Order header cache	On status change
order:status:{order_id}	Pub/Sub	N/A	WebSocket fanout	Ephemeral
drivers:geo:{city}	GEO sorted set	None	GEOADD driver locations	Updated each heartbeat
drivers:available:{city}	Set	None	Available driver UUIDs in a city	SADD/SREM on availability change
agent:{driver_id}:lock	String	30 s	Distributed lock during assignment (SET NX EX 30)	TTL only
location:order:{order_id}	Pub/Sub	N/A	Real-time location broadcasts to WebSocket subscribers	Ephemeral
idempotency:{key}	String	24 h	Idempotent response cache	TTL only
notif:{user_id}:unread	Counter	7 d	Unread notification badge	Decrement on read
geocode:{h3}	String	7 d	Reverse geocode cache	TTL only
ratelimit:{scope}:{key}	Counter	60 s	Sliding window rate limit	TTL only

7. Data Classification

7.1 PII

Encrypt at rest (column-level via KMS where listed); mask in logs; restrict to least-privileged roles.

Table	Columns
users	name, email, phone
addresses	contact_name, contact_phone, line1, line2
client_profiles	date_of_birth, gender
user_locations	latitude, longitude
driver_location_logs	latitude, longitude
otp_requests	phone, ip_address, otp_hash
sessions	ip_address, phone
audit_logs	ip_address
notifications	title, body, data (may include PII references)
restaurants	manager_phone

7.2 PCI-Adjacent

Never log; encrypt at rest + in transit; restricted to payment service only.

Table	Columns
payments	payment_token_encrypted

7.3 Sensitive Financial
KMS envelope encryption.

Table	Columns
restaurant_owner_profiles	fssai_license_number_encrypted, gstin_encrypted, pan_number_encrypted
driver_profiles	driving_license_number_encrypted, vehicle_registration_encrypted
bank_accounts	account_number_encrypted, ifsc_code
restaurants	fssai_license_encrypted, gstin_encrypted
payments	amount, amount_paid, gateway_order_id, gateway_payment_id
refunds	amount, gateway_refund_id
orders	subtotal, taxes, delivery_fee, discount, total_amount, delivery_address_snapshot
order_items	item_price_snapshot, line_total

7.4 Internal / Public
Internal — default for everything else. Public — restaurants.name/city/cuisine_types/rating, menu_items.* (excluding audit), driver_profiles.rating, total_deliveries.

8. Partitioning Strategy

Table	Strategy	Key	Retention
orders	Range (monthly)	created_at	Hot 6 months / Archive indefinite
order_items	Range (monthly)	inherits via order created_at	Same as orders
order_status_history	Range (monthly)	changed_at	Same as orders
driver_location_logs	Range (daily)	recorded_at	Hot 48 h / Cold 30 d / Delete
notifications	Range (monthly)	created_at	Hot 3 months / Archive 1 year
otp_requests	Range (daily)	created_at	7 days then purge
audit_logs	Range (monthly)	occurred_at	7 years (compliance)
sessions	None (moderate volume)	—	Purge inactive > 90 days

pg_partman schedules creation/drop; cold partitions are detached and shipped to S3 (Parquet) via a nightly job.

9. API-to-Table Mapping
100% coverage across the 49 LLD endpoints. Tables involved in audit_logs writes are listed where the API performs a writeable mutation.

#	Endpoint	Tables Read	Tables Written
1	POST /auth/check-phone	users	—
2	POST /auth/register	—	users, otp_requests, audit_logs
3	POST /auth/send-otp	users	otp_requests, audit_logs
4	POST /auth/verify-otp	otp_requests, users	otp_requests, users, sessions, audit_logs
5	POST /auth/logout	sessions	sessions, audit_logs
6	POST /onboarding/{role}/init	document_type_definitions	onboardings, onboarding_documents, client_profiles I restaurant_owner_profiles I driver_profiles, bank_accounts, addresses, audit_logs
7	GET /onboarding/{role}/status	onboardings, onboarding_documents	—
8	PATCH /onboarding/{id}/submit	onboardings, onboarding_documents	onboardings, audit_logs
9	POST /onboarding/{id}/resubmit	onboardings, onboarding_documents	onboardings, onboarding_documents, audit_logs

10	GET /users/me/profile	users, *_profiles, bank_accounts, addresses, restaurants	—
11	PUT /users/me/profile	users	users, client_profiles I driver_profiles, audit_logs
12	GET /users/me/addresses	addresses	—
13	POST /users/me/addresses	—	addresses, audit_logs
14	PUT /users/me/addresses/{id}	addresses	addresses, audit_logs
15	DELETE /users/me/addresses/{id}	addresses	addresses (soft delete), audit_logs
16	GET /restaurants	restaurants	—
17	GET /restaurants/{id}	restaurants	—
18	GET /restaurants/{id}/menu	menus, menu_categories, menu_items	—
19	POST /restaurants	users, restaurant_owner_profiles	restaurants, audit_logs
20	PUT /restaurants/{id}	restaurants	restaurants, audit_logs
21	POST /restaurants/{id}/menu	restaurants	menus, menu_categories, menu_items, audit_logs
22	POST /restaurants/{id}/menu/items	menu_categories	menu_items, audit_logs
23	PUT /restaurants/{id}/menu/items/{id}	menu_items	menu_items, audit_logs
24	DELETE /restaurants/{id}/menu/items/{id}	menu_items	menu_items (soft delete), audit_logs
25	POST /cart	menu_items, restaurants	carts, cart_items
26	GET /cart/{token}	carts, cart_items, menu_items	—
27	DELETE /cart/{token}/items/{id}	cart_items	cart_items
28	DELETE /cart/{token}	carts, cart_items	carts, cart_items
29	POST /orders/quote	carts, cart_items, menu_items, restaurants, addresses	—
30	POST /orders	carts, cart_items, menu_items, restaurants, addresses	orders, order_items, order_status_history, carts (status='converted'), audit_logs
31	GET /orders/{id}	orders, order_items, restaurants, addresses	—
32	GET /users/me/orders	orders, order_items, restaurants	—
33	POST /orders/{id}/cancel	orders, payments	orders, order_status_history, refunds, audit_logs
34	PUT /orders/{id}/status	orders	orders, order_status_history, audit_logs
35	GET /orders/{id}/tracking	orders, deliveries, users, driver_profiles, order_status_history	—
36	POST /payments/initiate	orders	payments, audit_logs
37	POST /payments/verify	payments	payments, orders, audit_logs
38	POST /notifications/send (internal)	—	notifications, notification_delivery_logs
39	GET /notifications/me	notifications	—
40	PUT /notifications/{id}/read	notifications	notifications
41	POST /delivery/assign (internal)	orders, restaurants, driver_profiles	delivery_requests, audit_logs
42	POST /delivery/partner/respond	delivery_requests	delivery_requests, deliveries, audit_logs
43	GET /delivery/{orderId}	deliveries, users, driver_profiles	—
44	PUT /delivery/{orderId}/status	deliveries	deliveries, delivery_status_history, audit_logs
45	POST /delivery/location	—	driver_location_logs (+ Redis GEOADD)

46	PUT /delivery/partner/availability	driver_profiles	driver_profiles (+ Redis SADD/SREM)
47	GET /delivery/partner/earnings	deliveries, orders	—
48	POST /users/me/location/resolve	—	user_locations
49	POST /uploads/presigned-url	—	— (S3 only)

10. Security at DB Layer

- Encryption at rest: AWS KMS-backed AES-256 for tablespaces (TDE / encrypted EBS) plus column-level envelope encryption for all *_encrypted columns. Keys rotated annually; kid stored alongside ciphertext.
- Encryption in transit: TLS 1.2+ enforced on PostgreSQL (ssl=on, sslmode=verify-full); client certificates for service identities.
- Row-Level Security (RLS): addresses, orders, notifications, carts, payments — policy user_id = current_setting('app.user_id')::uuid. restaurants, menus, menu_items — policy by owner_id (and managers via restaurant_managers mapping table). deliveries, delivery_requests, driver_location_logs — policy partner_id = current_setting('app.user_id')::uuid for driver role.
- Roles: app_rw (service runtime, no DDL), app_ro (analytics/replicas), payment_svc (only role with SELECT on payments.payment_token_encrypted), audit_writer (INSERT-only on audit_logs), dba (break-glass, MFA-gated).
- Backups: nightly logical (pg_dump --format=directory) plus 7-day PITR (WAL archive to S3, versioned bucket, MFA-delete). Cross-region replica.
- Audit: every write API writes a row to audit_logs (append-only, partitioned, 7-year retention). DB triggers auto-emit order_status_history and delivery_status_history.
- Hardening: log_statement='ddl', log_min_duration_statement=500ms, log_connections=on. PgBouncer in front; pg_hba.conf allows only VPC CIDRs.
- Secrets: rotation via Secrets Manager; no creds in app config.
- Schema migrations: forward-only via flyway/liquibase; reviewed PR + automated linting (no PII in plain text).
- PII access logging: views over PII columns (e.g., users_pii_view) emit access events to audit_logs.

Summary

Metric	Count
Total PostgreSQL tables	29 (28 + audit_logs + document_type_definitions – collapsed)
Total APIs covered	49
Tables with audit columns	15
Tables with soft delete	5
Partitioned tables	7 (orders, order_items, order_status_history, driver_location_logs, notifications, otp_requests, audit_logs)
Redis key patterns	16
Encrypted columns	10